

Internship Weekly Report

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From September 1, 2026 to September 7, 2026 (Week 1)

Assignments and Status:

Date	Major Assignments	Log Hour	Status
Day 1	- Software Testing Fundamentals: Studied the definition and purpose of software testing, and why testing is necessary within scope, time, and budget constraints. - Testing vs. Debugging vs. QA: Distinguished Testing (identifying defects — "Is there a problem?") from Debugging (fixing the cause of defects), and Testing (checks the product to find problems) from Quality Assurance (improves the process so fewer problems occur). - Error, Defect, Failure, Root Cause: Studied the chain from human error to defect to system failure and root cause. - The Seven ISTQB Testing Principles: Studied all seven principles and worked through five examples applying Principles 1, 3, 5, and 6, and the impossibility of exhaustive testing, to practical scenarios. Daily Report Link: Day 1.pdf	8.3 hr	Completed
Day 2	- SDLC Models: Studied the Waterfall model (sequential phases, testing after development), the V-Model (testing planned alongside development), and Agile/Scrum (iterative development, sprint cycles of 1–4 weeks, continuous testing) along with basic Continuous Integration (CI). - Test Levels: Studied Unit, Integration, System, and Acceptance Testing. - SDLC Questions: Completed 10 questions covering SDLC definitions, Waterfall, V-Model, Agile, sprint cycles, the four test levels, testing in Waterfall vs. Agile, CI, and SDLC model selection. - Practical (Student Registration App): Built an SDLC and test-level flow mapping Unit, Integration, System, and Acceptance Testing to the application's development stages. - Scenario Matching: Selected appropriate SDLC models for a banking compliance app (V-Model), a startup MVP (Agile), and IoT firmware updates (Waterfall/V-Model), with reasoning for each. Daily Report Link: Day 2.pdf	8.1 hr	Completed

Date	Major Assignments	Log Hour	Status
Day 3	- BRD vs. FRD vs. User Story: Studied the differences between a Business Requirements Document, a Functional Requirements Document, and a User Story. - Business, Functional, and UI Rules: Studied the distinction between the three rule types using an attendance-eligibility example. - Static Review Techniques: Studied Informal Review, Walkthrough, Technical Review, and Inspection, from least to most formal. - Practical (Mind Maps): Built a mind map for an NFC Attendance System (Student, Attendance, Reports branches), broke down a Login requirement into Username, Password, Login Button, and Error Handling, and built a Ticket App mind map (Ticket Lifecycle, Reporting, User Interface, Roles) with every requirement tagged as a Business, Functional, or UI Rule. Daily Report Link: Day 3.pdf	8.5 hr	Completed
Day 4	- Test Planning: Studied the components of an IEEE-style test plan, including objectives, scope, approach, entry/exit criteria, deliverables, tasks, environment, responsibilities, schedule, risks, and approvals. - Test Scope: Defined features to be tested for a Member Application (registration/login, password reset, profile management, subscriptions, renewals/cancellation, directory/search, notifications, admin panel, reports) and features excluded from scope (payment gateway internals, hosting provider uptime). - Test Strategy: Defined test levels (Component, Integration, System, UAT), test types (Functional, Usability, Security, Performance, Compatibility), and techniques (equivalence partitioning, boundary value analysis, decision tables, exploratory testing). - Entry/Exit and Suspension/Resumption Criteria: Defined conditions for starting testing, completing testing, and suspending/resuming testing after a blocking issue. - Practical (Member Application Test Plan): Prepared a complete IEEE-style test plan, including an eight-week testing schedule and a nine-item risk register (e.g., payment processing failure, exposure of member data, login failure under peak load), each rated by likelihood, impact, and business-impact justification with a mitigation action. Daily Report Link: Day 4.pdf	8.1 hr	Completed
Day 5	Week 1 Revision: Revised testing fundamentals, SDLC models and test levels, requirement analysis, Business/Functional/UI rules,	8hr	Completed

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	static review techniques, and test planning, with particular emphasis on SRS, Test Plan, and Test Strategy. • Test Estimation Techniques: Studied Work Breakdown Structure, Three-Point Estimation (optimistic, most likely, pessimistic), and the Function Point Method. • Selenium and Java: Selenium introduced as a future automation testing topic; Java installed in preparation for upcoming Selenium learning. • Practical: Used the DemoQA Book Store Application Login module as the practical context for connecting requirements, test planning, and testing concepts. Daily Report Link:Day 5.pdf		

Outcomes:

During Week 1, the internship built a foundation in manual software testing concepts, beginning with core terminology and the seven ISTQB testing principles, and the distinctions between testing, debugging, and quality assurance. This progressed into SDLC models — Waterfall, V-Model, and Agile/Scrum — and the four levels of testing, applied through 10 SDLC questions, a Student Registration Application test-level mapping, and a scenario-matching exercise across banking, startup, and IoT contexts. The week then moved into requirement analysis, covering BRD, FRD, and User Story formats, Business/Functional/UI rule classification, and static review techniques, applied through mind maps for an NFC Attendance System, a Login requirement breakdown, and a fully rule-tagged Ticket App. This was followed by test planning, where a complete IEEE-style test plan was prepared for a Member Application, including test scope, test strategy, entry and exit criteria, a schedule, and a nine-item risk register with business-impact justifications for each identified risk. The week concluded with a revision of SRS, Test Plan, and Test Strategy, an introduction to three test estimation techniques (Work Breakdown Structure, Three-Point Estimation, and Function Point Method), and preparation for future Selenium automation testing through the installation of Java, using the DemoQA Book Store Application Login module as practical context.

Conclusions:

Week 1 established a solid conceptual and practical foundation in manual software testing. The progression from testing fundamentals and SDLC models to requirement analysis and formal test planning provided both theoretical understanding and hands-on documentation experience, most notably through the Member Application test plan and its risk register. The closing revision and introduction to test estimation techniques consolidated the week's learning, while the installation of Java and the introduction of Selenium set the direction for automation testing to be taken up in the coming weeks.